Name: [Your Name]
Lab Program Number: 09
Date: [Submission Date]

# Title: Understanding and Implementing 2D Translation

## Theory

### *2D Translation*

2D Translation is a geometric transformation that moves every point of an object by the same distance in the x and y directions. It is used to reposition objects in a 2D coordinate system without altering their shape or orientation.

**Mathematical Representation:**
Given a point (x,y) and translation distances dx and dy, the new coordinates (x',y') are:

x' = x + dx
y' = y + dy

**Matrix Representation:**
In homogeneous coordinates, translation can be represented using a 3×3 transformation matrix:

[x' y' 1] = [1 0 dx; 0 1 dy; 0 0 1] · [x y 1]

**2D Translation Algorithm:**

1. Start
2. Input original coordinates (x,y) and translation vector (dx,dy)
3. Compute new coordinates: x' = x + dx, y' = y + dy
4. Update object position to (x',y')
5. Stop

### *Program for 2D Translation*

```
#include <stdio.h>
#include <graphics.h>

void drawTriangle(int x1, int y1, int x2, int y2, int x3, int y3) {
    line(x1, y1, x2, y2);
    line(x2, y2, x3, y3);
    line(x3, y3, x1, y1);
}

void translate(int *x1, int *y1, int *x2, int *y2, int *x3, int *y3,
               int dx, int dy) {
    *x1 += dx;    *y1 += dy;
    *x2 += dx;    *y2 += dy;
    *x3 += dx;    *y3 += dy;
}

int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");

    int x1 = 100, y1 = 100;
    int x2 = 200, y2 = 100;
```

```
    int x3 = 150, y3 = 200;

    // Draw original triangle
    drawTriangle(x1, y1, x2, y2, x3, y3);

    int dx = 50, dy = 50;

    // Translate triangle
    translate(&x1, &y1, &x2, &y2, &x3, &y3, dx, dy);

    // Draw translated triangle
    drawTriangle(x1, y1, x2, y2, x3, y3);

    getch();
    closegraph();
    return 0;
}
```

## Implementation / Output

**Output Description:**
The program displays a triangle before and after translation. The original triangle is drawn in one color, and the translated triangle is drawn in another. The output clearly labels:

**Object:** Original triangle
**Image:** Translated triangle
**Quadrant:** Both triangles are shown within the first quadrant.

(Attach a printout of your graphics output here, with clear labels.)

## Conclusion

This lab successfully demonstrated the implementation of 2D translation in Computer Graphics. By applying a translation vector (dx,dy), the triangle was shifted accurately across the coordinate plane. The exercise reinforced the mathematical foundation of translation and its practical application in graphical systems. The output clearly showed the movement of the object without any distortion, confirming the correctness of the implemented algorithm.

## References

https://drive.google.com/file/d/1TGVqe6k75SXQRoCdWrzK2Za6gfhrO-BY/view?usp=sharing